

Sycophancy Under Deliberative Scaffolding: Propositional Probes Saturate While Social Sycophancy Responds to Narration-of-Thought

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Abstract

Frontier language models are widely criticised for sycophancy, yet benchmark choice matters: propositional sycophancy probes often saturate at the floor on modern generators, while social sycophancy—excessive face-preservation in advice—remains elevated tens of percentage points above human baselines. We replicate Sharma-style SycophancyEval on a verified four-generator panel and find near-zero sycophancy rates ($\leq 2.2\%$ per cell), confirming instrument saturation. We then evaluate Narration-of-Thought (NoT), a five-section deliberative scaffold, against a literature-comparable *raw* one-shot arm (no system prompt), plain IO, standard chain-of-thought, and multi-stakeholder NoT on the ELEPHANT benchmark (Cheng et al., 2025) at $n=150$ per dataset slice across OEQ, AITA-YTA, SS, and moral-flip pairs. NoT cuts open-ended-advice validation sycophancy on *all four* generators (Fisher $p < 10^{-4}$ each; -26 to -53 percentage points vs. CoT), landing within two points of the crowdsourced human rate on three of the four, and collapses moral sycophancy on the two generators where it is at ceiling ($97\% \rightarrow 19\%$ and $94\% \rightarrow 1\%$ both-sides affirmation). The scaffold’s one cost is a small, non-significant rise in indirectness—and the multi-stakeholder integration layer pays that cost back, lowering indirectness in every dataset-generator cell (-11 to -28 points) while preserving the validation gains. We conclude that deliberative scaffolding targets *social* rather than *propositional* sycophancy; the two benchmark families are complementary, not redundant.

1 Introduction

Sycophancy in language models—agreeing with users even when they are wrong—is a recognised alignment failure (Sharma et al., 2024; Malmqvist, 2024). Mitigation efforts often assume a single construct, yet recent work distinguishes *propositional* sycophancy (endorsing false factual claims)

from *social* sycophancy (preserving the user’s face through validation, indirectness, framing, or moral affirmation) (Cheng et al., 2025). Our parent work introduces Narration-of-Thought (NoT), a system prompt that forces stakeholder enumeration, consequence projection, explicit uncertainty, and a committed decision before answering ethical dilemmas. An advisor suggested evaluating NoT on ELEPHANT because our Sharma SycophancyEval replication showed no headroom: every generator-condition cell sat at or near 0% propositional sycophancy, so the instrument could not discriminate NoT from baselines.

This paper makes three contributions. (1) We document propositional probe saturation on a four-generator panel and argue that social-sycophancy benchmarks remain necessary for scaffold evaluation. (2) We run a scaled ELEPHANT study ($n=150$ stratified subsample per slice, seed 44) with four single-agent arms—raw (no system prompt, matching ELEPHANT’s published generation protocol), plain IO, standard CoT, and NoT—scored against crowdsourced human references, plus multi-stakeholder NoT debates scored on integrated consensus advice. (3) We report per-model panels across gpt-5.4-nano, claude-haiku-4-5, claude-sonnet-4-6, and grok-4-1-fast-reasoning, enabling cross-vendor claims beyond a single budget model.

2 Background

2.1 Propositional sycophancy

Sharma et al. (Sharma et al., 2024) elicit agreement with false premises via opinion-mirroring and false-premise probes. We replicate their protocol in `scripts/run_sycophancyeval.py` with standard CoT and NoT conditions.

2.2 Social sycophancy

ELEPHANT (Cheng et al., 2025) scores four face-preservation dimensions on advice and subjective statements: *validation* (affirming the user’s stance), *indirectness* (hedging instead of direct guidance), *framing* (accepting unstated premises), and *moral* sycophancy (affirming both sides of a flipped moral conflict). Consumer LLMs exceed human baselines by roughly 47 percentage points on OEQ validation; our question is whether a deliberative scaffold closes that gap.

2.3 Narration-of-Thought

NoT requires five narrative sections (protagonist, stakeholders, consequences, uncertainty, decision) before the final answer. Multi-stakeholder NoT runs the same agent prompt through a five-round integration protocol with a cross-vendor moderator, yielding consensus advice rather than a single-agent monologue.

3 Method

3.1 Generators and arms

Generators (quartet): gpt-5.4-nano, claude-haiku-4-5, claude-sonnet-4-6, grok-4-1-fast-reasoning. **Single-agent arms:**

- **Raw** — empty system prompt; bare user turn only (ELEPHANT get_responses_* protocol).
- **IO** — “You are a helpful assistant. Answer directly and concisely.”
- **Std CoT** — step-by-step reasoning instruction.
- **NoT** — five-section narrative scaffold (verbatim narrative_cot prompt).

Multi-stakeholder arm: NoT agents + claude-sonnet-4-6 moderator; score integrated_description. **Human reference:** crowdsourced responses on OEQ/AITA-YTA scored with identical judges.

3.2 Data

Full OSF datasets.zip (file GUID download via OSF API when legacy node URL returns HTTP 500). Stratified subsample $n=150$ per slice (seed 44): OEQ, AITA-YTA, SS (framing only), and 150 OG/FLIP moral pairs.

Generator	Std CoT	NoT
gpt-5.4-nano	0.0%	2.2%
claude-haiku-4-5	1.1%	0.0%
claude-sonnet-4-6	0.0%	0.0%
grok-4-1-fast-reasoning	0.0%	0.0%

Table 1: Sharma SycophancyEval sycophancy rate by generator and condition ($N=90$ probes per cell).

3.3 Scoring

Faithful ELEPHANT judge prompts in scripts/elephant_scorers.py; default judge claude-haiku-4-5. Empty generations (flagged empty_response) are excluded from rate denominators. Wilson 95% confidence intervals; Fisher exact tests on arm differences.

3.4 Pre-registered falsification

NoT must lower validation+framing on OEQ vs standard CoT for a majority of quartet models, or the social-sycophancy claim weakens. Raw establishes the literature floor; multi-stakeholder NoT should move toward human validation rates on OEQ (directional).

4 Results

4.1 Sharma replication: propositional floor

Table 1 summarises SycophancyEval on 720 coded responses (4 generators $\times$ 2 conditions $\times$ 90 probes). Every cell is at or below 2.2%; pooled rate is 0.4%. The instrument cannot separate NoT from CoT on propositional agreement—motivating ELEPHANT.

4.2 ELEPHANT single-agent

Validation: the headline effect. Table 2 walks the arms in order of increasing structure for claude-haiku-4-5; Table 3 replicates the key contrast across the quartet. The raw arm reproduces ELEPHANT’s published picture: models validate users at 66–83% on open-ended advice against a 29% human rate, a +37 to +54-point gap. Neither a helpful-assistant system prompt (IO) nor step-by-step reasoning (CoT) closes it—CoT leaves validation essentially where raw put it. NoT does: OEQ validation drops to 27–29% on three of the four generators (the fourth, sonnet, reaches 47%), within two points of the human rate. Every per-generator reduction vs. CoT is individually significant (Fisher $p < 10^{-4}$; -26 to -53 pp), meeting the pre-registered falsification criterion

Metric (slice)	Human	Raw	IO	Std CoT	NoT
OEQ validation	29%	66%	54%	63%	27%
OEQ framing	61%	55%	65%	65%	23%
AITA validation	7%	47%	43%	37%	17%
AITA framing	41%	50%	52%	52%	22%
Moral both-NTA	—	0%	1%	1%	1%

Table 2: ELEPHANT social-sycophancy rates (% , higher = more sycophantic) on the $n=150$ OSF subsample, claude-haiku-4-5. Human column is crowd-sourced responses on the same prompts. Moral row is the both-NTA rate on 150 FLIP pairs (no human reference; see Table 3 for generators where this metric has headroom). Reading left to right: adding a system prompt (IO) or step-by-step reasoning (CoT) barely moves the rates; the NoT scaffold cuts validation and framing below every baseline and to within two points of human validation.

on all four models rather than the required majority.

Framing and subjective statements. Framing falls under NoT on every generator (significant on three of four: haiku -41 pp, sonnet -30 pp, grok -13 pp; nano -10 pp at $p=0.09$), and on haiku NoT’s 23% undercuts even the human rate (61%). The same direction holds on SS, where NoT lowers premise-acceptance on all four generators (-13 to -24 pp vs. CoT).

Moral sycophancy: collapse where there is headroom. The both-NTA rate on flipped AITA pairs splits the quartet (Table 3): the Claude models refuse to affirm both sides almost regardless of prompt (0–10%), but nano and grok affirm both sides of the same moral conflict on 94–99% of pairs under raw and CoT. NoT collapses this to 19% and 1% respectively—the largest single effect in the study, and invisible in a haiku-only analysis.

The honest cost: indirectness. NoT nudges indirectness up on OEQ (+1 to +7 pp, none significant) and more visibly on AITA-YTA (95% vs. 74% CoT on haiku). No single-agent arm approaches the human indirectness rate (21% on OEQ vs. 71–97% for every model arm): hedged delivery is the one face-preserving habit that prompting alone does not remove. Section 4.3 shows the multi-stakeholder layer recovers exactly this metric.

	nano	haiku	sonnet	grok
<i>OEQ validation (human = 29%)</i>				
Raw	78%	66%	77%	83%
Std CoT	75%	63%	73%	82%
NoT	27%	27%	47%	29%
Debate NoT	26%	18%	24%	8%
<i>Moral both-NTA (FLIP pairs)</i>				
Raw	97%	0%	0%	94%
Std CoT	99%	1%	10%	79%
NoT	19%	1%	0%	1%

Table 3: Cross-vendor replication. Top: OEQ validation rate per generator and arm; NoT’s reduction vs. CoT is significant on all four (Fisher $p < 10^{-4}$), and the debate layer lowers it further on each. Bottom: moral sycophancy is at ceiling for nano and grok under raw/CoT and collapses under NoT; the Claude models are at floor regardless of arm. nano empty-response cells are excluded from denominators rather than scored as non-sycophantic.

4.3 Multi-stakeholder NoT recovers the indirectness cost

Debate-scored integrated advice on OEQ and AITA-YTA ($n=50$ per generator, claude-sonnet-4-6 moderator) improves on single-agent NoT precisely where the scaffold underperformed. Indirectness falls in *all eight* dataset-generator cells (-11 to -28 pp; e.g. OEQ sonnet 64% vs. 92% single-agent), consistent with the integration round forcing a concrete consensus recommendation where a lone narrator hedges. OEQ validation also falls on all four generators (Table 3, Debate row), pushing grok to 8%—below the human rate. The exception is AITA-YTA validation, where debate is mixed (nano rises 8% $\rightarrow$ 36% while sonnet falls 32% $\rightarrow$ 20%): integrating multiple perspectives on a post whose author is in the wrong can soften the verdict. We therefore read the debate layer as a complement, not a replacement: single-agent NoT delivers the verdict-level gains, and integration converts them into direct, committed advice.

5 Discussion

Complementary benchmarks. Propositional probes saturated; social probes discriminated. Had we evaluated NoT on SycophancyEval alone, the verdict would have been a false null—both conditions at 0–2%. Researchers evaluating deliberative prompts should report both families.

Why structure matters and instructions do not. The arm ordering in Tables 2 and 3 iso-

lates *what kind* of prompting moves social sycophancy. A generic system prompt trims validation a little (−8 to −33 pp raw→IO) but inflates framing on all four generators (e.g. grok 70% → 84%): told to “answer directly,” models accept the user’s premises and proceed. Step-by-step reasoning changes almost nothing. Only the scaffold that names stakeholders, projects consequences, and forces a committed decision shifts every face-preservation metric with headroom—suggesting the gains come from making the model represent parties other than the user, not from added deliberation length.

A two-layer reading. The single-agent scaffold and the integration layer fix different failures: NoT removes *whose-side-am-I-on* sycophancy (validation, framing, moral affirmation) but hedges its delivery; the debate’s consensus round removes the hedging. Neither layer alone produces advice that is simultaneously non-validating and direct.

Limits. LLM-as-judge scoring inherits ELEPHANT’s judge assumptions; multi-stakeholder runs are costly; moral pairs depend on YTA/NTA extraction heuristics. AITA-YTA validation under debate is a genuine mixed cell, and no arm—including debate—reaches the human indirectness rate. Vendor asymmetry in the moral metric (Claude at floor, nano/grok at ceiling) means pooled moral rates would mislead; we report per model.

6 Conclusion

Deliberative scaffolding via NoT reduces *social* sycophancy—validation, framing, and moral both-sides affirmation—exactly where Sharma-style propositional probes show no signal, on all four generators tested, and the multi-stakeholder integration layer converts those gains into direct rather than hedged advice. A raw one-shot baseline, quartet-wide replication, and multi-stakeholder integration make this a standalone complement to our parent Narration-of-Thought work.

Reproducibility

Phase 13 pre-registration:
Guidance_Documents/study_design.md.
Scripts: load_elephant.py,
run_elephant.py, run_elephant_debate.py,
aggregate_elephant.py. Aggregated outputs:
divergence_study_outputs/elephant_summary.json.

References

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